

Active Inference BookStream #001.01 ~ "Governing Continuous Transformation"

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foreign it's December 6 2022 we're in active bookstream number 001.01 a numbering convention that will be explained soon it's a discussion on the book governing continuous transformation by vijayan kazri and I'll pass the blue hi everyone I guess that's my job is to explain the uh bookstream numbering so this is our first book stream that's why it's bookstream001 and this is our first dot zero contextualization video for this bookstream series so each week we will present a chapter ish a chapter or so of the book and then periodically we will get together for the zero zero one dot one zero zero one dot two and so forth those discussion sections so if you want to participate either in a contextualization video or in a discussion video please get in touch with us and we would love to have you um on the stream and I'm blue I've been here before so hopefully you've seen me before and know me by now and I'm an independent researcher in New Mexico and I will pass it off to Tyler first time appearance I think in the active inference Institute so Tyler do you want to introduce yourself yeah cool thanks for having me this is my first time here to actually inference so I have a an unusual background actually don't come from in an academic background at all a lot of it's been about applied governance so I actually started as a strategy consultant working in major corporations more recently it's gotten more to Dow governance I currently lead product and protocol design for the 4K protocol it's a project that is connecting the physical world to the on-chain world and I'm also a contributor at medigov where I think about how we can model these abstract formalisms about governance in a real world setting and so a lot of times when I think about Dow governance but most recently it's been thinking about active inference and implicit implications for real world settings so we are a participatory online lab that is communicating learning and practicing applied active inference these are all of our media links um and we will be recording And archiving this live stream for future reference um everyone is welcome and hopefully we will be following good etiquette for live streams and not arguing or talking over each other for the duration of the book streams um okay so this is

our first book stream um our goal is to learn and discuss this textbook governing continuous transformation reframing the strategy governance conversation it's a 2022 book by Bijan kesri and this video like always is just an introduction to some of the ideas not a review or a final word at all so we will just go through here um the sections of this book stream we're going to do the book map and the road map some keywords the preface part one and chapter one will be covered in today's video and in the coming weeks we will be continuing to discuss this so get in touch if you want to partake in these conversations um so I guess we introduced ourselves briefly um oh backwards so if either you guys want to maybe mention something you're excited about or that you liked or remembered from the book from these first sessions um otherwise we can go on to discover it I think we have a lot of material Tyler what are you most excited about yeah I mean I think for me like especially being in like real world organizations non-profits a lot of times we think of like uncertainty as something that could be wished away or something that we're incrementally going to solve like oh if we only were collecting like more information we would be in a better spot or I don't think we had more capabilities this we wouldn't have this problem um but a lot of times we kind of get very anxious about that uncertainty rather than we're an active inference and free energy governance that uncertainty is placed at the center and it's something you could actually take ownership over but also be a source of you know creative uh growth and so I think that's really exciting and inspirational and yeah really excited to be here today cool I'm excited too to unify some of the concepts of active inference with governance which I know a little bit about but not very much and now that we're like formalizing our structure as an Institutional entity in the active inference Institute um I'm excited to learn more about governance and how things should work um okay so this book has four parts so we're here well we're not even in part one yet but we're going to go over to the beginning of part one today so part one is reframing the strategy governance conversation part two is about free energy governance and part three is a conclusion for the book and then there are some interviews in part four along with acknowledgments and an epilogue and this is from the bookmap software so this is like our road map for a book um but this bookmap software is used in in trading so if you guys have never seen it that's what these graphs are from so when I think about bookmap I always think about that uh okay so our roadmap for today so we're going to get into the preface there are no section headings so we just kind of chunked out like what makes sense um in terms of of the text and how it reads so we're going to go over a summary of the fep how to ground firms in the framework of active inference strategy governance model models versus meta governance

the triplet structure of free energy governance and then realists and anti-realists in management science and then just a brief intro to part one and then chapter one which is pretty short thinking born of curiosity revolt and change um so here's like a huge list of pre-words of pre-words of keywords that we're going to get into today and I think Tyler is going to start us off um in we're in section one in the preface in the summary of the free energy principle Tyler's gonna start us off talking about the fep Tyler and mute sorry uh so the pre-energy principle or fep as we're going to be calling it uh this is John's definition that he gives in the preface so the fep uh is a mathematical formulation that explains from first principles to characteristics of biological systems that are able to resist Decay and persist over time it rests on the idea that all biological systems instantiate a hierarchical generative model of the world that personally minimizes its internal entropy by minimizing free energy blew anything you'd want to add there no I think um it's best like we could take a deep dive into the fvp there are many live streams where that's done so if you want to learn more about the free energy principle I would just refer out to other live streams it's best here to take what Bijan presents the aspect that he's using and go forward from there cool and I think uh similarly we'll do a similar cursory treatment of Markov blankets uh and so markup blankets so we could think about it is that it's a statistical formulation for defining the internal versus the external so the definition that he gives is Marco blankets establish a conditional Independence between internal and external states that renders the inside open to the renders the inside open to the outside but only in a conditional sense so I.E the internal states only see the external States through the veil of the Marco Blanca again huge topic we're not going to go super into it unless you're gonna add anything blue no it's great um and and there are many many live streams like really if you want to get into it probably the emperor's new Markov blankets is maybe where we undertake like a very thorough um discussion of the Markov blanket if you want more information there all right so grounding firms in the framework of active inference so here's a claim from the book learning to actively infer will determine the firm's competitive advantage and survival in a discontinuous and distributed world the more fluid the boundaries between firm and Eco Niche the more important the definition of what constitutes a system and how non-discipitative markovian boundaries preserve the firm's Integrity I.E belief systems our interest is Shifting away from pursuing deterministic truth through clever analysis towards designing entailing a generative self-organizing model that empowers the firm to master the shifting conditional independencies between the known and unknown and what lies within and Beyond its control whoa so there's a lot there so I think these next keyword slides

hopefully will unpack it a little bit uh active inference um which we've seen many times in the active inference Institute is a process Theory grounded in the free energy principle that explains Behavior perception planning and action in terms of probabilistic inference um Fijian doesn't yet mention active inference but he talks about learning to actively infer which I'm going to actively infer means active inference and here we see this classic action perception Loop where we have internal States and external States separated by a Markov blanket or a non-disruptive markovian boundary and we have sensory input and action and these are the ways in which an agent can interact with this world do either you guys have anything to add here Daniel um in that framing I found it really interesting when there's increased fluidity or fuzziness or blurriness of the engagement of the firm with the niche or the context or the community in those cases it's like more important to have a uncertainty oriented definition of what the thing is and that is what confronts every organization today yeah interesting um okay so he talked a little bit about shifting away from deterministic truth and so like I I like Googled because I'm a truth person I'm weird about truth if you've ever like listened to me talk about it before so deterministic truth is like the truth um and I I liked this paper like I just Googled it like what do people talk about what is deterministic truth um and I I found this paper approaching determinist deterministic and probabilistic Truth a unified account and and in this paper it's a 2021 paper so pretty new um in this paper they talked about the truth itself like the truth being either deterministic like I think about that like hitting a bullseye like the truth is the number seven and then probabilistic truth is more like a probability distribution around that truth like hitting seven plus or minus the standard deviation or like a probability density right um and this paper talked about like deterministic and probabilistic Truth like that the truth is one of those things and then also there are ways to approach it like you're approaching your search of the truth in either a deterministic or probabilistic way so I thought that this was an interesting read and kind of if you are interested in delving deeper into like deterministic deterministic versus probabilistic and these theories and um approaches and and Truth likeness I thought this was a cool um read yeah I think one thing oh sorry no the whole thing to add there is just I think like our personal ideas about what truth are like whether it's terministic versus a probabilistic conception of Truth really carry over onto the organization level and so whatever kind of delusions you have about your personal truth in your personal life a lot of times they view the organization as like the CEO abuse the organization as an extension of their own self and so their own world view gets carried over and you know strapped on to the organization of the world view um and so it's really interesting that you

could kind of frame these like philosophical ideas about truth that are often applied at an individual level and then kind of like scale that up into an organization yeah very cool um and we see that in this um this generative self-organizing model again this is kind of like it was cool to just search up what is a self-organizing model um and and we see generative Model A lot um we don't really see self-organizing model we see generative model in terms of self-organization but a generative self-organizing model like what is this really um um and so just this is just from this um Stanford machine learning cheat sheet supervised learning cheat sheet so here it says discriminative models divide the data space into classes by learning the boundaries and this is like the different neural networks whereas generative models understand how the data is embedded into the space like an auto encoder or boltzmann machine or self-organizing Maps so this is like a self-organizing model and you can really see here like this is the like a decision boundary on the left in the discriminative model and like either it's a cat or a dog right and on the right it's there's a generative model like it's a cat with some degree of certainty or it's a dog with some degree of of certainty and so you see these like probability distributions like probably it's a cat but then there's like some fuzzy like if your data point lands like a little bit in the red or a little bit in the blue like then then it might be a cat or it might be a dog maybe we don't have enough information to tell um so I thought that this was a cool like replication of the probabilistic truth versus deterministic truth like either you're looking for cat or dog or you're looking for with what probability is this a cat or a dog um do you guys have anything to add here cool okay so this is moving on out of this claim into the next claim um The Firm is envisioned as an inactive sophisticated and resource dependent inference machine in form of a generative learning model essentially it is a statistical model linking in circular causality action perception firm resources and the unknown external world and I just put this active inference Loop up here again so you have like the firm and the world the firm would be the agent and then you have the resources of the firm which would be like the internal States and then the external world which is everything that's not the firm um and I highlighted these terms inactive and sophisticated because there's a lot to unpack here um and I think it's kind of in the textbook maybe like just you can say the words but unless you like open them up and look what they mean inside look inside of them like you don't have any idea the kind of depth that goes behind these things so um do you guys have anything to add there on that okay um inactivism is a position in cognitive science that argues that cognition arises through a dynamic interaction between an acting organism and its environment and so these are the four e's of uh cognitive science and so or cognition and they're embodied and active

extended and in cultured so cognition is grounded in our senses and concrete experience that's embodied enacted means it's goal directed action in the real world extended is cognitive systems include the tools the devices the people around us our smartphone these kinds of things and embedded or that's kind of what we always refer to as in cultured but I think that they kind of mean the same thing so this is cognition is woven into culture and learning is done in a social context so these are the the embodied So when you say the firm is inactive um yeah that that's it it's good for goal directed it's doing goal directed action in the real world and active and sophisticated or do you guys have anything to add there on the 40s okay sophisticated is from this paper I believe a sophisticated inference by Kristen and colleagues and we've done sophisticated affective inference I think on a live stream but I don't think we've ever done sophisticated inference but the affective inference is um like a later paper and covers this one a lot so sophisticated inference is really like in deep time right so here I'll just read this the figure caption so I have this like decision tree model a picture from the paper and it says this schematic summarizes the accumulation of expected free energy over paths and trajectories into the future this can be construed as a deep tree search where the tree branches of reliable actions at each point point in time and the likely outcomes consequent upon each action so this is one two three four time steps here and going off so this is like a trajectory the path of least action uh so we talk about you know the way that a system can decide and go forward in in space so that's what's meant by sophisticated I think oh Daniel yeah the purpose of introducing these terms from cognitive sciences and Neuroscience like the 4E cognition and of this type of model it's not a novel idea to make a decision tree that's how for example early chess playing artificial intelligence's work it's the way that the expected free energy can be calculated across these different Paths of consideration with respect to a generative model that are allowing new ways of thinking about these kinds of deep time and nested strategic questions um okay so temporal thickness and counterfactual depths here in this claim Bijan says the proposal on offer here is that the Mind comes into being when self-evidencing has a temporal thickness models can be thicker and thinner deeper or shallower depending on how far forward they predict or counterfactual depth how far back they post it that is whether they can capture how things might have ended up if they had acted differently there is no real reason for Minds to exist they appear to do so simply because existence itself is the endpoint of a process of reasoning Consciousness I'd contend is nothing grander than inference about my future and in this this is from bhajan's book but he's quoting friston 2017 paper um so that temporal thickness goes back to this sophistication right which is what why like so here you

can see how far back you can post it like if I had done something differently what could have happened um this counterfactual reasoning and then and prediction going forward into the future okay going on to strategy governance models versus meta governance all right so I'm kind of out of my uh Comforts I'm here so I might call in Tyler to help me out a little bit um but Bijan says the common understanding underlying entrenched strategy governance models rather erroneously has been that action is a consequence of perception and therefore strategy must be environment driven so I look up strategy governance model and it gave me like some figures and this is one of them and it appears to be um pretty linear but but there's not a lot of feedback happening with the external environment like there's Vision the strategy a road map design and then insights come forward to the design and back to the division and then there's like a performance review so I don't know Tyler do you want to give any additional comments here on this yeah I mean I think something that could be potentially confusing for people who are maybe not coming from an academic background is that like a lot of times what Business Leaders are thinking about how they're going to plan for the action that their organizations to take what the strategy looks like you know they're getting data right and they're having feedback from the bottom levels of the organization feedback to the top levels right and so that they're taking information they're creating a perception of that and then they're creating a strategy based on that that kind of in a linear fashion the organization executes and at some point they reflect back and be like hey did this work or not what do we do with our new information right but really what they're not realizing in the process of that is like in that linear flow they're not recognizing how they're constraining the perceptions and actions of their organization in a top-down fashion and allowing and allowing bottom-up information to change how information and actions can be changed in the future so like that kind of cyclical nature really to understand it's really more conceived in this linear way of like hey we're going to collect as much information as we can our information is going to be imperfect but we're going to get better over time rather than like hey fundamentally our actions and perceptions are constrained by the actions we're taking right now so it's like just a different model for thinking but it's a lot more linear and not doesn't have that cyclical causality uh baked into it cool and I think about here like um it seems like it's very like um sequential like one after another after another after another whereas like the active inference kind of method like you have to sometimes act in order to perceive right like the if you want to know whether like the surface is rough or smooth you have to move over the surface with your finger to determine if it's rough or smooth so it's action and perception that are happening

simultaneously and I think that that that is the shift that Bijan is calling for um as he says yeah and I think just to add in that like a little bit is that I think a lot of times Business Leaders conceive of like reality as like independently existing out there and it's just like a limitation of their tools and resources to understand reality rather than like the nature of perception itself of being able to understand reality and so I think that's really like the fundamental difference and a lot of times people don't realize that that's even like relevant at all uh and so that's why this is fun to talk about that's for sure why like what is deterministic truth like where is that reality um and it all kind of links together um so Jean goes on to say in fact action upon the firm's Eco Niche akin to pointing the light torch in the dark determines perception inference is inactive the generative model itself never encodes anything it just expresses the circular causality among all four states the generative model hence is a model of the firm's econish if not it is no good regulator and consequently The Firm will rather sooner than later dissipate in the face of entropic forces um and so I included here A different kind of strategy governance model that's like the Lean Startup model that's like a little bit more action and perception happening in like a more simultaneous way I don't know if you want to add anything here Tyler yeah I think um a lot of the way we talk about like Lean Startup model of really the least our model is basically like you don't necessarily know what people want and also it's not that you're discovering what people want is that you're putting ideas out there and finding how people respond respond to them uh and so it's basically truth is something that you create you can create a new state of the world rather than like uncovering it so it's kind of like a method for uncovering something that's going to work and a lot of times it's kind of done how can like very like fluffy and also just like experiential terms of like hey this works we've done this we built this company this way but what's interesting about active inferences that can take that very like fluffy wispy framework and apply that into a more rigorous context uh so yeah like a lot of these ideas that we're talking about now like they're not necessarily like new and like especially like startup world but there hasn't been great language for talking about it cool all right so here's another claim therefore the firm's true governance challenge is a matter of meta governing its generative model as an enacted sophisticated and resource-dependent belief system exercising the entire organization quote the body itself in Contours we should now start seeing a model emerging that underwrites a form of Auto autopoiesis I.E self-organization effectively it is all about casting inferences optimization mathematically by way of gradient descent starting with two essential constituents first a hyper prior which is the firm's resistance to entropy in order to persist and two epistemic foraging and hypothesis

testing I.E acting upon the Eco Niche to generate belief evidencing data um do you guys have any comments here do you want to delve into this a little more Daniel yeah one interesting uh usage of a term there is that the firm's true governance challenge is meta governance and that's a term that is used differently or compatibly but with a different focus in medigov and so just to kind of Flesh that out here meta governance is being used in the sense of metacognition like self-reflection memory anticipation attention planning all of these individual level cognitive and metacognitive processes that we model in active inference often medicov in another sense is meant like social cognition like relationships amongst organizations or amongst teams so like within a team you have their own governance structure then also there's teams of teams and that's a meta governance structure with respect to the team and then that's very interesting because the two times that we've seen nested models nested generative models inactive inference are the metacognition models and then the ecosystem nested models so it really brings Integrity between thinking about what happens like within and across groups cool thank you um okay so hyper prior um in Bayesian statistics a high proprietor is a prior Distribution on a hyper parameter that is on a parameter of a prior distribution as with the term hyper parameter the use of hyper is to distinguish it from a prior distribution of a parameter of the model for the underlying system so like the hyper parameters influence the prior here and like the hyper parameters here influence the parameters here and so it's it's backwards at least one step in time in this sophisticated format so there's oh yeah this is a better diagnosis or illustration the hyper prior goes to the prior goes to the data so it's just in a previous time step in some way um anything to add here all right epistemic foraging I pulled this from active inference and learning this this quote um it says inactive inference Behavior has explorative epistemic and exploitative pragmatic aspects that are sensitive to ambiguity and risk respectively where epistemic Ambu ambiguity resolving Behavior enables pragmatic reward seeking Behavior and the subsequent emergence of habits um and and I just wanted to kind of point out here that that active inference is kind of like has been Illustrated as like a balance a way to balance exploration and exploitation and epistemic and pragmatic value um and like in business is this like knowledge and and money is that like epistemic and pragmatic value like knowledge of your customers your clients so it's just trying to think of like how this might um translate to that like like money pragmatic values like more research sources and you know like epistemic value is like more knowledge of your client base or or of of the future maybe um uncertainty production and I could just give like a concrete example of that because like it was especially true in software engineering where that is actually frame and explicitly these

terms and uh like growth organizations so if you're on the growth team you're like okay we can either like turn up the dial on this thing that's working we're like okay like we know this flow is working this wizard for this checkout flow is working let's make it like 10 faster would be an example of exploitation where like we know that's going to work in some level where acceleration could be something we have like really no idea this is like a shot in the dark let's figure out the minimum amount of effort we can put out to to uncover new territory and so like this exact trade-off is very much explicitly thought of in a lot of tech organizations very cool thank you yeah so epistemic foraging is like you know searching around so like you know for for epistemic seeking out um epistemic value um okay so moving on to the triplet structure of free energy governance uh Tyler this is for you I think um so you know Jean's frames like really there's like a key question of your energy governance when you're thinking about your organization how do you embody the academic emergency um and it's really about how you create and update your generative model so that the firm can survive and he presents this framework of thinking about it and he calls it the triplet structure of re-energy governance there is structure cognition and then capabilities so we'll go through uh each of these in turn all right structure so oh sorry this is confusing B that header is wrong just get to the next slide um so this John writes that connectivity emerges in flow and language bottom up stimuli emanating from resource markets such as clients Technologies suppliers another are only as relevant and meaningful to Resource configuration asset allocation as clearly as top-down predictions and prediction models articulated communicated and above all programmed to process bottom up incoming signals um so like we kind of touched on this earlier but like a lot of times decision making at organization is very linear where like at the very top you're coming up with a strategy and then like that's executed in current at every rung of the hierarchy and a lot of times like okrs if you've ever heard of this it's like a really good example of this it's like objective key results where like there's an objective that the top the organization comes with and then like the minions the next run have their sub subsequent parts of that tree and it kind of filters down to the rest of the um where sometimes what's not as clearly articulated is about the interconnections between different parts of the hierarchy and how those different parts of the stack uh have to relate to each other so like a really good example of this is that you know this is very true in tech companies tech companies collect a ton of data and none of that data is like ever processed in a meaningful way and so like a lot of times you'll say like oh like we got we're data driven company we collect all this information but in reality like a lot of information is trapped at the bottom rungs of the organization it may not have like

the political ability or also even the technical ability to process that in a meaningful way and bubble that up through the organization and a lot of times there's a huge gap in being able to actually process that information and bubble it up appropriately uh in a way that can actually like crater the organization and so this is like designing this in a really thoughtful way is much harder than it would seem and even if you call yourself a data driven organization there's a good chance you're not processing data in a way that can actually be generate new top-down models later on so yeah anything to add there where Virginia all right let's go to the next one cognition all right so cognition Embraces the environment as enacted through relations and interactions constructed in our brains environment is not objectively accurate independently exists but is revealed by a firm's actions and perceptions that those firms that those very actions generate in fact the consequences of the firm's activity construct it's equal niche um and so we've already kind of touched on some of the earlier slides But the A lot of times these firms conceive of external reality existing independently of their own actions and perceptions and then it's just like up to them to get better tools and information um to be able to like understand that reality like understanding this fog of war that they're like slowly uncovering but they actually do real insights then also the actions that they're able to take so like to make this like a little bit more concrete you can say like I mean if you're not this very much applies in a real Sense on like a website where like if you're not collecting information on views on your page in a very basic level it's really hard to like generate actual information out of that right and so you have to basically make a decision of like this is the information we want to gather this is what we care about and that's going to structure the insights you can glean in the future and like what you can do with that information in a similar way like if you're not putting resources out for interacting with the world like creating a new Department that's maybe a little bit more exploratory that's going to constrain the actions you can take in the future so all this is saying is that uh the reality doesn't exist independently of you it's something that you actively perceive through your actions and perceptions down blue and add anything there is it like actively perceived or actively create um actively sorry I actively create a better way of saying it well or or both I mean I think like you have to be um either actively perceiving but but um I don't know I think it's it's a rare treat when someone takes responsibility or even like a company takes responsibility for um the reality that they're creating um and like we see this especially like in tech companies like I don't know Facebook and I mean like just like a lot of the technology has been life-changing and world changing so like they're really actively creating reality Daniel yeah this is uh interesting about the revealing

of the firm's generative model through its actions and we've talked about that four-fold partition with the internal external States and then the sense and action blanket States so the firm has access to sense and internal states to some extent but the world or others have access to one's actions and or inference on their consequences in the niche and so it's a way to talk about what the entire action perception Loop is with all of its different inward facing and outward facing Dynamics yeah to add a little more here too like there's a lot of focus on around this idea as well and like startup culture and so I I'm a totally butcher there's like a Steve Jobs Club totally butcher about something around the lines of like if you had done a marketing survey of people before they made the iPod and asked them how they want to listen to music they wouldn't have told you that they wanted an iPod they would have told you something else that was conditioned based on their previous experiences products and so the idea is that you just put something out in the world it's a creative act and that you create something in the world and then the world responds to it um and so there's like this big idea especially in like Tech culture around like reality is something that you will into existence rather than um discover independently yeah it's interesting too to like think about I mean especially you think about Google Glass that came out like I don't know 10 years ago or something now I'm probably not that long ago but it's been a little while and I think like now there's like the re-release of like Google Glass or some kind of similar product like the world just wasn't ready when the Google blast version came out but but like we'll be ready soon we've they've wielded into existence and it'll be that way soon all right capabilities the last one is capabilities um so the inferring firm is constantly optimizing belief propagation as well as updating beliefs uh sophistication that process that process of optimizing recognition densities is only as good as the firm's General the world model so we are no longer just interested in the actions consequences per se uh about how the actions consequences impact ulterior beliefs another way of saying this is that the firm has to actively balance you know this exploiting for reverse um exploring and I think there's like a lot of again folk wisdom and especially in Silicon Valley culture around this trade-off whereas like a lot of times we say hey failure is good as long as you're learning and integrating those those insights um and so yeah it blew anything to add there no good all right um okay so realists and anti-realists in management science all right so here's a claim from Bijan management science is equally divided along not too dissimilar realists and anti-realist fault lines it is best captured by Carl week 1977 when he states that environment is located in the mind of the actor and is imposed by him on experience in order to make that experience more meaningful it's seldom Dawns on

organizational theorists to look for environments inside the heads rather than outside of them indeed the rise of cognitive lens in strategy management and governance research has enriched our understanding around improving the design of strategy governance processes but has largely remained constrained by the notion that environment is something objective out there waiting to be analyzed and adapted to um and in in this section uh Bijan talks about Blitz scaling and um mentions some like examples of of this anti-realist uh claim so he says Google Uber Airbnb and Facebook are the product of a fundamentally anti-realist enactment of an interaction with the respective Eco niches powered by unique and continuously evolving sets of capabilities fostering action generated top-down bottom-up prediction error minimization to iteratively optimize their very generative models and implied recognition posterior probabilities of beliefs about the actions consequences these companies redesign while they fly which is in the words of Silicon Valley veteran Reed Hoffman equivalent to throwing yourself off a cliff and assembling your airplane on the way down Sullivan 2016. in fact we are more in control than we resist to believe Blitz scaling has been integral to the founding DNA of these super outperformers they listen to incense what wants to emerge they are the enacting designers of their very Eco Niche for these companies strategy is rather a matter of ex post sense making then ex-anti long-term planning it is a matter of survival and I just thought that that was maybe a good example of this anti-realism um thing that that he was talking about uh do you guys understand an idea that kind of arose as you were talking about this is that one interesting point about Google is that they're also kind of famous for trying really hard to do this but failing to create new products so they've like created and killed like hundreds of different new like products that people genuinely like truly loved even that were quite Innovative and so it's kind of interesting of like how you even think about like what is trying to emerge and whether that's best done in like a single monolithic organization versus in a more like decentralized distributed way um so I don't know I this has been like a tension I've been kind of feeling throughout the book as well it's kind of like what is it that this entity is trying to exist or it needs to exist or is it something that is more distributed and decentralized is more appropriate in certain cases so having loved and lost several Google things in the past that have like people are either uh yeah Google or Google App Maker like the low code app developing app development thing I mean just like tons of things right having loved and lost I wonder if this is like epistemic exploration on the part of Google really so so if like you're you know I don't know I feel like people come in and like I'm gonna build this app and Google gives them a lot of freedom and they're like go ahead build the app or build the platform or

build the thing and do the thing and they build it and it's great and people love it and then they move on and do something else and then nobody steps up to support the love of this part or what was this person's like child at Google right so I feel like maybe that might be a lot of what's happening but but perhaps in this like creation and then unsupporting creation and unsupporting Google is continuously generating like epistemic knowledge right it's epistemic foraging of and learning about the niche and what people like and what they will use you know it it like on the grand scheme well I think it's I think that's true but I'm not going to make this argument but a lot of people have made the argument that what they should have been doing the whole time is exploiting because like their home business is just ads and they're like just like run that until you die like that's the only thing that matters like your resources aren't well used here by like trying to do like all of these random things um and so like there is like a legitimate argument too of like the limits to which you can explore maybe I don't know that's probably out of scope for our discussion today but um all right Daniel yeah just one short note the entire discussion of epistemic and pragmatic value is motivated motivated by the fact that they're unified in the free energy functions that are used to determine perception cognition and action in an integrated way so at the very least conversation can be had about adaptive changes in these variables at different levels of the organization in a different if not better way that's very integrated with how we think about the world internally and built upon Neuroscience in that area as well as about thinking about the complex Dynamics and the relational dynamics of the niche which for information doesn't exist until you're seeking it out and in feedback with it and inattention of your previous data in anticipation of future data all of these cognitive Dynamics so the anti-realist take doesn't need to be someone who doesn't think like a real world exists or anything like that even though that's like too deep for this current discussion it's just a purely pragmatic approach to thinking about and modeling these kinds of scenarios very cool all right what else ah okay so here we are we got through the preface that was the preface um and so getting into part one of the book uh there's just this small quote which I'll read the it says reframing the strategy governance conversation the evidence suggests that we are in the midst of an evolutionary punctuation we are witnessing a mass extinction in the corporate world in the early Decades of the 21st century since 2000 52 of the Fortune 500 companies have either been acquired merged or have declared bankruptcy it is estimated that 40 of the companies in existence today will shutter their operations in the next 10 years merely following the trends of change is not enough just like organisms facing the great oxidation event organizations need to reinvent the way they interact with the changing world and this is

a quote from digital transformation survive and thrive in an era of mass extinction at 2019 by siebel all right chapter one thinking born of curiosity revolt and change all right here's our abstract Tyler I'm gonna let you read this one all right environmental complexity and velocity have been dominant moderators and strategy and governance research the underlying structure of environmental change is becoming increasingly distributed and discontinuous the disciplinating nature of Information Technology is challenging the trench business models as well as the Logics of organizing socialized top-down strategy processes are rather anti-clocked unless dynamically and generatively intertwined in real time with bottom-up stimuli and data emanating with field operations in pursuit of prediction error minimization neither established Logics of organizing or prevalent corporate governance models are a match for wholesale digital reinvention and continuous strategic renewal nice and going on he says in fact akin to the AI Evolution from a supervised to a deep learning logic corporate governance is challenged to evolve from a supervised and rule-based best practices Learning System to one of unsupervised unlearning I.E self-organization to survive the firm must be conceived as one cross hierarchically integrated inference machine organizing is generative prediction error minimization strategy is redefined as top-down bottom-up predictions processing the search for knowledge is nourished not nourished by certainty it is nourished by a radical absence of certainty thinking born of curiosity revolt and change and that's a quote from roveli 2021. all right all right Tyler so uh one of the core premises of the app the abstract is saying hey the environmental complexity and velocity increasing over time and that makes it more challenging for organizations um to survive so what he says is have been dominant moderators and strategy and governance research indeed the rate of change has not only been accelerating but the underlying structure of change has become discontinuous sense of urgency and heed full interrelating are critical to survival in a world that is now commonly perceived as uh full of surprises and so yeah this is something just by being on like the inside of a lot of these organizations including corporations a lot of times it's perceived as hey there's it's a Scary World out there we just need to get our information better to kind of understand what's actually happening so we don't have um any problems but I think the problem is that like the world is changing so much faster than it had been before where like this kind of Relentless approach of hey we're just going to get more we're going to gradually improve our processes to wash away this uncertainty that doesn't work as well anymore and I think this kind of former approach uh strategy is very much encapsulated by like Six Sigma design I don't know if you guys are familiar with that but it's kind of like a manufacturing idea of like hey we're

just going to slowly and incrementally increase the Precision of our techniques and our processes because these any instruments you have Cascades across your whole supply chain and all of your other processes so we just need to figure everything out and so everything will be perfect and we won't have any issues um and that may be like somewhat of a illusion that you can tell yourself if things aren't changing that quickly and maybe that'll work for like some constrained amount of time but in the modern world it's hard really hard to have that illusion you have to have a different mental model of the world um all right move on to the unless blue and Daniel had anything no all right um so the second piece of this is like yes the world is changing really quickly and that means that there's going to be more discontinuities and so he says our discounting of these are fundamentally a function of the cost of interacting with the outer World in form of generating and testing predictions and so Wix is not discontinuity it's really about like this gap between perceiving the world and saying okay how are we going to take action how are we going to update our generative model accordingly accordingly and what he's saying is that you know as the world becomes more complex change is much faster than ever did before there are more of these discontinuities but also the discontinuities are larger there's just more information to integrate and so like on one level you could say like this is really terrifying world's changing what do we do but he also sounds this optimistic note of saying like well yes the modern world creates more discontinuities but it also creates more tools for navigating business continuities and so the first one he says is well you know is about distance remediation and decentralization so it's a lot easier to aggregate data and process that information and have backup plans in case plan a fails you have plans you know B through Z instead the other aspect of this is AI and so it's a lot easier to process bigger and deeper sets of data and so he's trying to sound this optimistic note that hey not All Is Lost there is a way to navigate through this uncertainty I just have to say I love this meme as someone as someone who's like tried to like organize and restructure my life probably I don't know 500 000 times at this point um like it's like do I just do the work or do I spend time like agonizing about how to like document and structure and like time plan my day into five minute blocks so like like I invest so much time in like some method right and then I'm just like okay well I methodized it and then I never used the method I just go back to doing the work so it's it's like this weird like all this time analyzing what like or dumping it into like moderating my efficiency is like wasting my efficiency to connect that to the earlier discussion about the two senses of meta this time cost associated with strategy and with consequences in the real world that time cost can be understood as a cognitive cost and something internal

that's happening metacognitive how should I decide how to decide how to perceive this some of those are variables and knobs you can understand and change other ones are really subliminal and then also that time cost strategy Nexus of course exists for organizations and teams and all of these nested and complex interactions so we want to use active inference as an integrative principle because it helps us connect what's happening on either side of the Markov blanket which might be like a data set passed between two areas of a community or of an organization understanding how the emission model and the reception model work together yeah to attach that a little more is that like you know I was I've been a product Analyst at uh some like a couple of different startups and you actually see this in a real way where like sometimes you can get an okay answer by just taking like the like a super like simple average of a couple different categories or you could try to do something really clever and create like an app like you know an ml model or something over um to predict something and a lot of times it's like it doesn't the predictive power you gain doesn't match the time cost of that and by the time you created this really sophisticated like publishable uh model the world has fundamentally changed so much that your model just like doesn't matter anymore uh and so like you're very like time-bounded and like how you can even like use the information you gather and the results you you glean so for example if you even if you have like a a checkout flow and you say oh yeah like uh with this button the checkout flow is 10 more effective than with this button well it could it will very likely be the change in six months the context of that checkout flow has changed so much that it's actually just kind of meaningless information and so like the time-boundedness of the information you collect and the perceptions you can form of that matter a lot um all right so the inference machine um so to survive the firm must become inference machine that continuously builds predictions which are calling to be bottom of challenge um and so I think this very much aligns with what we talked about before like agile methodology this kind of Lean Startup method of you build see if it worked you kind of like integrate those um those perceptions and learn from it use that to ideate and something that is a creative act into the world not necessarily something that is fully informed by data um and then you build and act on it um and he uses this quote a few times both in the premise I think he actually opens the book with this uh he says there's a Carlo ravelli quote the search for knowledge is not nourished by certainty it is nourished by radical absence of certainty thinking born of curiosity revolt and change so this is actually very important because I've actually encountered this tension very often in a lot of organizations where like a lot of organization is one of the data driven and they don't actually another way to say this is this is

the best product advice that was ever given any it was from CEO of my last companies and said there's two types of product work there's art and there's science and the science work is like you're like hey like we know this works we know we can speed up this checkout flow it might get like incrementally better we kind of can just keep going on there's like the exploit path but then there's art which is really just like not informed by the data and a lot of times organizations have trouble with this because it is truly a creative act and they're not comfortable taking that leap of faith and that's really what like you have to do to do anything uh meaningful and a lot of people are much more comfortable in the exploit path which is a lot easier to sell to your bosses it's a lot more comfortable and you have to really like CR like very consciously create a culture within your firm that like the explore path is like even okay and something that can be rewarded all right anything that you guys want to add here before we on the next slide all right um so something that I found of uh something I found that was interesting with you know reading this preface and the first chapter was that like a lot of the themes are kind of emerging in a less technical Frameworks and also kind of like almost like pseudo-spiritual ideas about business management uh and so this is like a quote that was from the paper uh or from the book it says that firms are called to open up all four channels of listening listen from what you know what surprises you from the whole and what and what you and what you sense once you emerge the emerging home right and so if we kind of think about integrating information across different levels of the hierarchy like uh Jean was talking about like you can think about that from a more like rigorous academic perspective but there's like a lot of other books especially that come over the last five years that are conceiving um you know almost like a pseudo-spiritual lens where it's conceiving the organization not as like the extension of the CEO's ego but really as like this entity that needs to be nurtured and cultivated and understood on its own terms one of the most famous books about this is called Reinventing organizations and this reading organizations introduce the idea of teal organizations these are like non-hierarchical heterogeneous um so in that book he says instead of trying to predict and control the future members of the organization are invited to listen in and understand what the organization wants to become what purpose it wants to serve which like on first reading like that can feel very like pseudo-spiritual and kind of like hand wavy but is actually like a very real thing that can actually be modeled in a more rigorous way with activator and so I think that's really exciting another way that this has been talked about has been uh in Brian J Robertson's book holography holocracy is another similar to like teal organizations it is a non-hierarchical form of governance and so he says each tension human

being sense is a signpost telling us how the organization could evolve to better express its purpose all right anything you guys want to add that cool so I don't know anything else that you guys want to add um just thanks for being here and going over this uh stuff with me um and is there anything else you're looking forward to or or what's next I'm really excited awesome oh just logistically how is the series going to continue sure so we'll finish one chapter a week is the goal um maybe like we're trying to still work out the logistics of the holiday but I think it looks consistent um so one chapter of the book per week something about that something like along those lines and then a discussion after each like section of the book so there's part one part two part three um and we will get in touch with the author and see maybe if he is willing to come and discuss some of these ideas with us or and hopefully we'll have other um people that want to participate in the discussions of the works Tyler and then I'll give a thought yeah I mean I think the thing I'm looking forward to is you know so far what I've been excited about is that putting a a concrete framework behind ideas that have so far been expressed in very abstract kind of hand wavy terms across you know different kind of startup folklore and um kind of business type businessy books and so I'm curious as a Next Step about how it can become a little more like tangible and practical and like for someone who is from the perspective like being in an organization how do you how would you actually design an organization to embody pre-energy governance um and so I'm really excited to see how that can become a little more concrete and clear cool the part I'm looking forward to is continuing to explore that explore exploit dialectic both of those in the extreme lead to failure in some way or at least increase risk and so this frames our strategizing as being like in some sort of landscape with explore and exploit which of course is really complex with these nested organizations and then you brought in the Art and Science or um science and engineering or just art and communication or whatever other dialectic and then even within an artist's day there's realities like accounting and the mixing of the paint and even the prompt engineering for someone who's using those kinds of artistic forms so it lets there be sort of a light and adaptive quote explore exploit scenario happening within the context of something that's a lot more meaningful which is the integration of like art science personal meaning Collective meaning and so on without denying that those kinds of explore exploit attentional and strategic Dynamics must happen also cool well thanks for participating guys and um we'll see you next time bye y'all